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SECTION 1

MAGNET SITING, INSTALLATION & COMMISSIONING OVERVIEW

TABLE 1-1
MAGNET SITING, INSTALLATION & COMMISSIONING
FLOW CHART & CHECK LIST

SITING & INSTALLATION			
Responsibility	Function	Reference Document	Complete
GE Installation Specialist & Shield Room Vendor	Magnet Room Requirements: • Safety (Exhaust & Venting) • Magnet Anchoring • Power & cooling • Mounting Plate installed (if required)	Direction 2243897, OpenSpeed Delivery & Installation Direction 2268297, OpenSpeed Mounting Plate Installation	☐
GE Installation Specialist	Preinstallation Check List Completed	Direction 2241391, Signa OpenSpeed Pre-Installation, Section 9, Pre-Installation Checklist Direction 2402692, Signa OpenSpeed EXCITE Pre-Installation, Section 9, Pre-Installation Checklist	☐
GE Field Engineer & Rigger	Site Delivery Review: • Access & Route • Clearances	Direction 2243897, OpenSpeed Delivery & Installation, Section 6, Moving Magnet To Mr Suite	☐
Rigger	Magnet Delivery	Direction 2243897, OpenSpeed Delivery & Installation, Section 3, Domestic Magnet Delivery, and/or Section 4, International Magnet Delivery	☐
GE Field Engineer	Magnet Component Checks: • Physical & Electrical	Direction 2243897, OpenSpeed Delivery & Installation, Section 5, Magnet System Component Check (by GE Service)	☐
Rigger	Move Magnet to MR Suite	Direction 2243897, OpenSpeed Delivery & Installation, Section 6, Moving Magnet To Mr Suite	☐
Rigger	Leveling & Bolting Down Magnet	Direction 2243897, OpenSpeed Delivery & Installation, Section 7, Magnet Mounting	☐
Continue with "Commissioning" Phase			

SECTION 1 - MAGNET SITING, INSTALLATION & COMMISSIONING OVERVIEW (continued)

TABLE 1-1 (CONTINUED)
MAGNET SITING, INSTALLATION & COMMISSIONING
FLOW CHART & CHECK LIST

COMMISSIONING			
Responsibility	Function	Reference Document	Complete
	From "Siting & Installation" Phase		
GE Field Engineer	MR Magnet Safety Document 2301164PRE Review & Compliance	Direction 2231934, SAFETY REQUIREMENTS Tab	☐
GE Field Engineer	Magnet Conversion to Operating Configuration; Immediate Vent Installation	Direction 2231934, SET-UP AND CALIBRATION, Section 1, Magnet System Installation	☐
GE Field Engineer	Magnet System Components Installation & Functional Checks: • Cryocooler • MRU • Magnet Monitor	Direction 2231934, SET-UP AND CALIBRATION, Section 1, Magnet System Installation	☐
GE Field Engineer	Magnet Commissioning Checks	Direction 2231934, SET-UP AND CALIBRATION, Section 2, Magnet Commissioning Checks	☐
GE Field Engineer	Connect Magnet Monitor	Direction 2230681, GEMS Magnet Monitoring Hardware (SHEM) Installation	☐
GE Field Engineer	Magnet Precool & Fill	Direction 2231934, SET-UP AND CALIBRATION, Section 3, Helium Precool and Fill	☐
GE Field Engineer	Ramp Magnet	Direction 2231934, SET-UP AND CALIBRATION, Section 4, Ramp Fill	☐
GE Field Engineer	Shim Magnet	Direction 2250758-1, <i>Signa OpenSpeed Service Methods</i> , or Direction 2402212-1, <i>Signa EXCITE OpenSpeed Service Methods</i>	☐
GE Field Engineer	Main Field Adjustment A/R	Direction 2231934, SET-UP AND CALIBRATION, Section 9, Field Adjustment After Shimming	☐
	Hand Off to Systems Calibration		

SECTION 2 - PRE-DELIVERY INSTRUCTIONS



Make sure ALL site requirements / conditions identified for the magnet SIGNA OpenSpeed Pre-Installation manuals (Direction 2241391 for non-EXCITE systems and Direction 2402692 for EXCITE systems), are met before scheduling magnet delivery. This will prevent installation delays, cryogen loss and the resultant ongoing magnet quenches, potential damage, environmentally-related problems and increased costs.

Note

- The latest version of Direction 2241391, Signa OpenSpeed Pre-Installation, (for non-EXCITE systems) can be found online at:

<http://3.87.118.28/optec3/common/07tdocs/2241391/2241391.htm>

- The latest version of Direction 2402692, Signa OpenSpeed EXCITE Pre-Installation, (for EXCITE systems) can be found online at:

<http://3.87.118.28/optec3/common/2402692/2402692.htm>

1. Visit the magnet site with rigging foreman to plan the move before the magnet is delivered.
2. Caution the rigger that the magnet is fragile. Sudden jolts can damage the magnet. Make riggers aware of the cost of a magnet replacement. This usually promotes more care while handling before the magnet is delivered.



Magnet has enclosures installed and enclosures cannot be pushed upon as damage to the enclosure may result.

3. Make sure all roads and paths leading to the Magnet Room are level and free from obstacles and holes. Rigger will be required to construct platforms where needed.
4. If roller dollies are to be used, have rigger bring steel plates to place along delivery route. Materials to be supplied by the rigger are specified in Section 3-2, Equipment / Tools, for domestic deliveries and in Section 4-2, Equipment / Tools, for international (air / sea) deliveries.
5. Make sure the Magnet Mounting Plate is installed on the floor before delivery. Refer to architectural drawings plus Direction 2241391, Signa OpenSpeed Pre-Installation, (for non-EXCITE systems) or Direction 2402692, Signa OpenSpeed EXCITE Pre-Installation, (for EXCITE systems).

SECTION 2 - PRE-DELIVERY INSTRUCTIONS (continued)

6. Make sure all equipment anchors are located, installed and tested.



Review guidelines with Shipper / Riggers prior to transporting the magnet.

Shipping and handling guidelines are provided in Tables 3-1, 3-2 and 3-3 for domestic deliveries and in Tables 4-1, 4-2 and 4-3 for international (air / sea) deliveries.

These guidelines must be followed to prevent any potential damage to the magnet during shipping and handling.

SECTION 3 - DOMESTIC MAGNET DELIVERY

WARNING!

LOSS OF MAGNET VACUUM WILL RESULT IN THE VIOLENT EXPULSION OF HELIUM GAS, WHICH CAN CAUSE ASPHYXIATION IN ENCLOSED AREAS. USE EXTREME CAUTION TO NOT CONTACT / DAMAGE THE VACUUM VESSEL DURING MAGNET TRANSIT OR SITING.

3-1 IN-TRANSIT HANDLING

TABLE 3-1
LIFTING / HANDLING INFORMATION

Magnet	Maximum Weight*	Maximum Tilt**	Forklift Capability
GE 0.7T	22,000 lbs. (9,980 kg.)	30 deg.	Only with pallet attached
* Weight includes magnet, poleface, helium, motorshields, gradient coils with cables, hoses, etc. and shipping cage. ** Maximum tilt angle off vertical axis is allowed only when supported by feet for forklift / rigging operations or moving on inclines (base lift).			

CAUTION

Make sure conditions specified in Tables 3-2 and 3-3 are met to prevent internal damage to magnet. The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

TABLE 3-2
SHIPPING INFORMATION

Magnet	Allowable Shipping Modes	Shipping Capability	Maximum Transit Time
GE 0.7T	Airplane*, Air ride only trailer, Boat or ocean going ship*	Warm or Cold	Cryogen refill for cold shipment ~ 30 days (to eliminate magnet precool requirement)
* Only for crated magnets. See Section 4, International (Air / Sea) Magnet Delivery.			



3-1 IN-TRANSIT HANDLING (continued)

TABLE 3-3
MAXIMUM IMPACT FORCE

Magnet	Maximum Longitudinal Force	Maximum Lateral Force	Maximum Vertical Force
GE 0.7T	1.2 G	1.5 G	1.5 G

Note

Only tie downs to magnet ring and lifting rails are permitted.

3-2 EQUIPMENT / TOOLS**3-2-1 Crane Lift**

TABLE 3-4
CRANE REQUIREMENTS

Item	Qty.	Capacity / Range	Furnished By:	Intended Use:
Crane	1	22,000 lbs. (9,980 Kg)	Rigger	Unloading or moving magnet (without pallet)
Spreader Beam	1	22,000 lbs. (9,980 Kg) Total Load Additional data shown in Illustration 3-3	Rigger	Unloading or moving magnet (without pallet)
Straps or Cables	2	22,000 lbs. (9,980 Kg) Total Load Additional data shown in Illustration 3-3	Rigger	Unloading or moving magnet (without pallet)
Sling (Strap or Cable)	2	2,000 lbs. (910 Kg) work load limit, 6 ft. (1830 mm) min. length	Rigger	Cage removal
Shackle or hook	2	2,000 lbs. (910 Kg) work load limit	Rigger	Cage removal
Wrench	1	Size: 9/16 in.	Rigger	Cage side removal



3-2 EQUIPMENT / TOOLS (continued)**3-2-2 Miscellaneous Equipment / Tools**

TABLE 3-5
MISCELLANEOUS EQUIPMENT / TOOLS FOR MOVING AND CHECKING MAGNET

Item	Qty.	Equipment / Tool Requirement	Responsible	Function
Magnet System Components Checks	1	Direction 2243897, GE 0.7T OpenSpeed Magnet Delivery and Installation	GE / SYS	Confirm electrical check values
Hydraulic or Toe Jack	4	Must support one end of magnet on 2 jacks or both ends of magnet on 4 jacks Carrying capacity: 12,000 lbs. (5,450 Kg) per jack	Rigger	Raise up magnet for roller dollies or leveling plates
Roller Dollies	2	Carrying capacity: 6,000 lbs. (2,730 Kg) per dolly	Rigger	Moving magnet to Magnet Room
Come-Along	1	Pulling capacity: 12,000 lbs. (5,450 Kg)	Rigger	Pull magnet into position
Level	1	24 - 36 in. (600 - 1000 mm) long	Rigger	Check magnet level
Torque Wrench and Socket	1	Torque range = 58 to 258 ft. lbs. Deep socket, size dependant on anchor selection	Rigger	Bolt down magnet
Socket Wrench and Sockets	1	3/4" socket size (add sockets for crate removal)	Rigger	Remove 1/2 in. crate lag screws
Portable Temperature Meter (P/N 2125073)	1	N / A	GE / FE	Check internal temperatures
Digital Volt / Ohm Meter	1	Fluke model 87 or equivalent	GE / FE	Perform magnet electrical checks
Wood Timber	2	4 in. x 6 in. x 8 ft. (100 x 200 x 2000 mm)	Rigger	Moving / positioning magnet to Magnet Room
Steel or Aluminum Plate	6	.25 in. x 18 in. x 6 ft. (7 x 500 x 1800 mm)	Rigger	Moving magnet to Magnet Room
Pry-Bar	2	Length as required to move magnet	Rigger	Moving / positioning magnet in Magnet Room
RuO Temperature Meter (P/N 2171219)	1	N / A	GE	Check second stage & Recondenser temperatures
Adapter Cable for Diode Temperature Meter (P/N 46-301620P1)	1	N / A	GE	Check internal temperatures
Circular or Chain Saw	AR	N / A	Rigger	For cutting wood timbers
Heavy Duty Drill & Drill Bits	AR	N / A	Rigger	To add holes to wood timbers
Min. .50 Inch Dia. Mounting Hardware	AR	N / A	Rigger	For attaching wood timbers to rails
Socket or Wrench	1	15/16 in.	Rigger	For removing lifting beams



3-3 IN-TRANSIT SERVICE

Cryogen depletion occurs in approximately 3 days after coldhead power is removed from a full magnet. The Magnet warms up to liquid nitrogen temperature (77 K) in approximately 5.5 days. If both coldheads are off for an extended period of time (> 5 days) due to transit, storage or power outage, refill of the Magnet to keep it cold should be considered to eliminate the need to precool it before filling later. Perform helium fill in conformance with magnet service manual Direction 2231934, SET-UP AND CALIBRATION, Section 3-4, Liquid Helium Fill.

WARNING!

THIS PROCEDURE MUST BE PERFORMED BY QUALIFIED PERSONNEL ONLY. MAKE SURE THE SAFETY PRECAUTIONS LISTED BELOW ARE COMPLETELY FOLLOWED WHEN PERFORMING ANY SERVICE THAT WILL RESULT IN HELIUM EXHAUST FROM THE MAGNET:

- **LARGE QUANTITIES OF HELIUM GAS WILL DISPLACE OXYGEN IN A CLOSED ROOM. GOOD VENTILATION IS REQUIRED FOR YOUR PERSONAL SAFETY. MAKE SURE THE MAGNET STORAGE AREA IS ADEQUATELY VENTILATED BY OPENING DOORS AND WINDOWS. THE ADDITION OF PORTABLE FANS MAY BE NECESSARY IN CASES OF A SMALL-SIZE ROOM WITH POOR VENTILATION.**
- **SKIN CONTACT WITH LIQUID CRYOGENS WILL CAUSE BURNS. WEAR PROTECTIVE CLOTHING, GLOVES (NONABSORBENT MATERIAL) AND GOGGLES WHEN TRANSFERRING CRYOGENS. MAKE SURE A SECOND PERSON IS PRESENT WHILE SERVICING THE MAGNET IN CASE OF ANY EMERGENCY.**
- **SMOKING IS PROHIBITED IN THE MAGNET AND CRYOGEN STORAGE ROOMS. LIQUID CRYOGENS CAN LIQUIFY ATMOSPHERIC OXYGEN THUS PRODUCING A HIGHLY ENRICHED OXYGEN LIQUID, WHICH IS HIGHLY COMBUSTABLE.**

Note

Magnet must be lifted in conformance with requirements shown in Illustrations 3-2 and 3-3 to avoid damage to external components from lifting straps / chains.



■ 3-4 CAGE REMOVAL



Use caution while lifting or sliding protective cage from magnet to avoid serious damage to the magnet components.

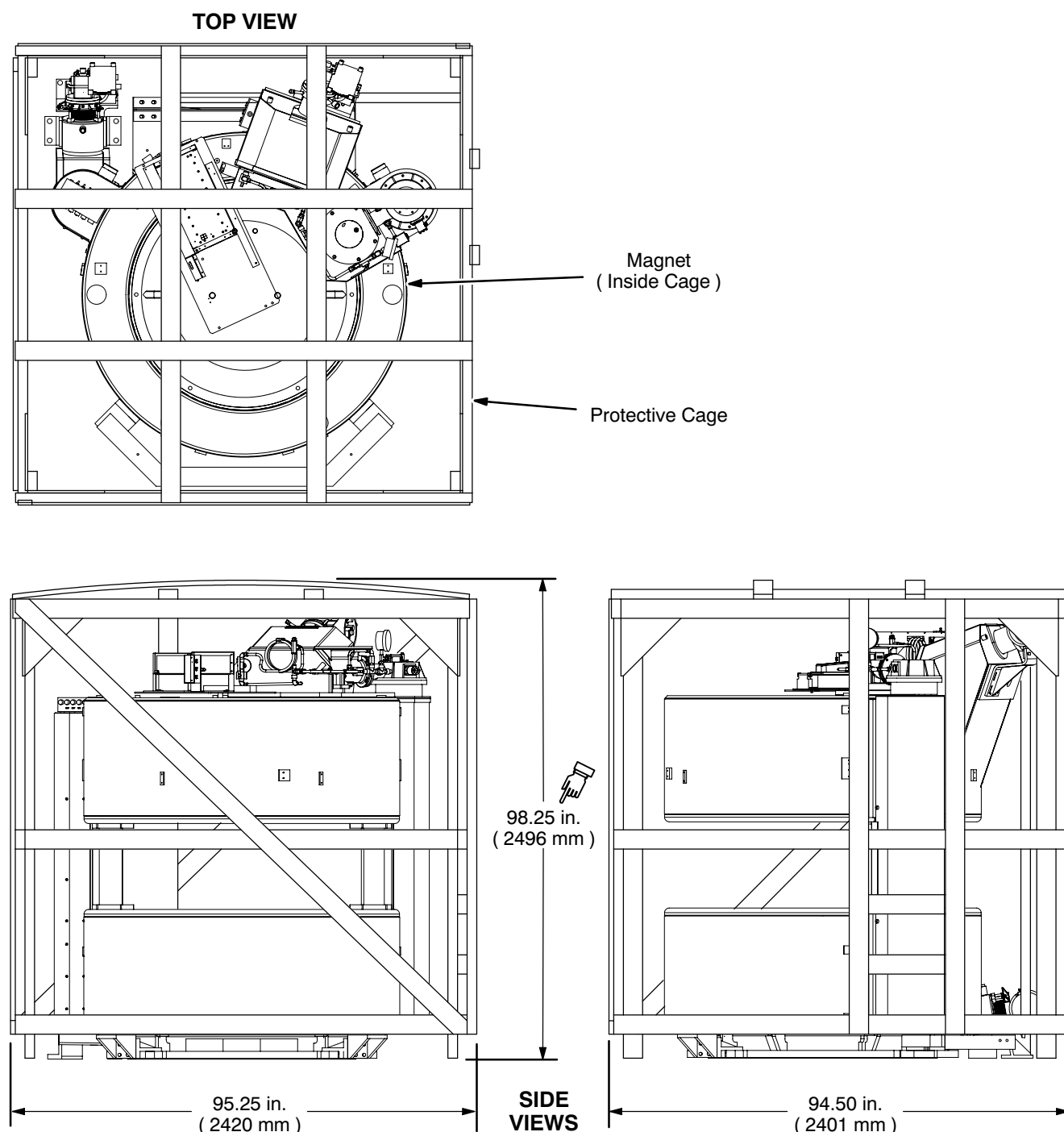
Note

■ Magnet must be lifted in conformance with requirements shown in Illustrations 3-2 and 3-3 to avoid damage to external components from lifting straps / chains.

- 1. Loop a lifting strap capable of lifting a minimum 300 pound (136 kg) weight through the cage's top frame and securely attach it to crane. See Illustrations 3-3 and 3-1.
- 2. Remove lag screws and frame from side of cage. Lift cage a minimum of 24 inches (609.6 mm), carefully sliding it away from the magnet at that height, or lift cage a minimum of 24 inches (609.9 mm) above top of magnet, swinging it away to remove. Be careful not to contact magnet with cage during this operation. See Illustrations 3-2, 3-3 and 3-1.



3-4 CAGE REMOVAL (continued)



PROTECTIVE CAGE / MAGNET
ILLUSTRATION 3-1

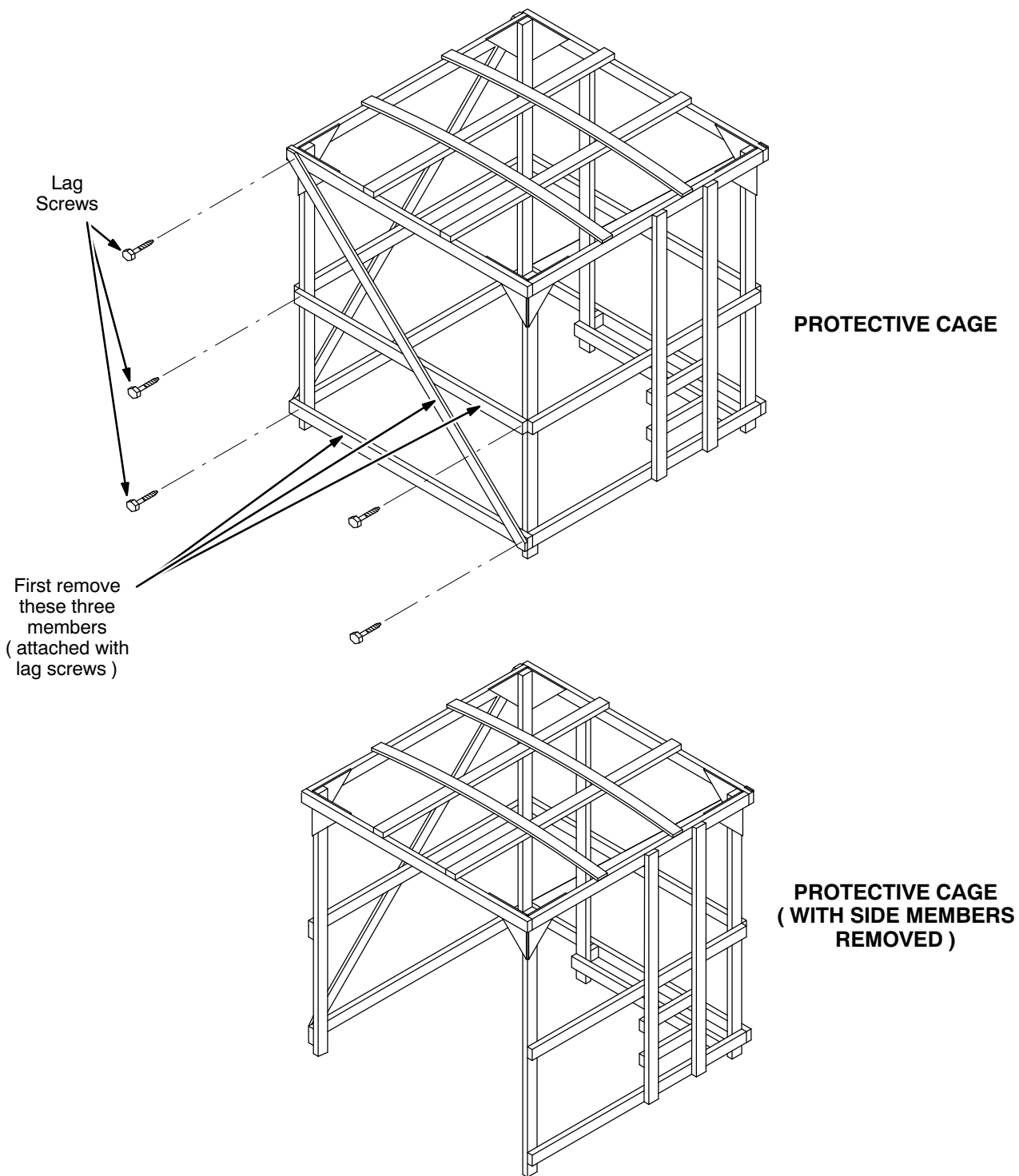
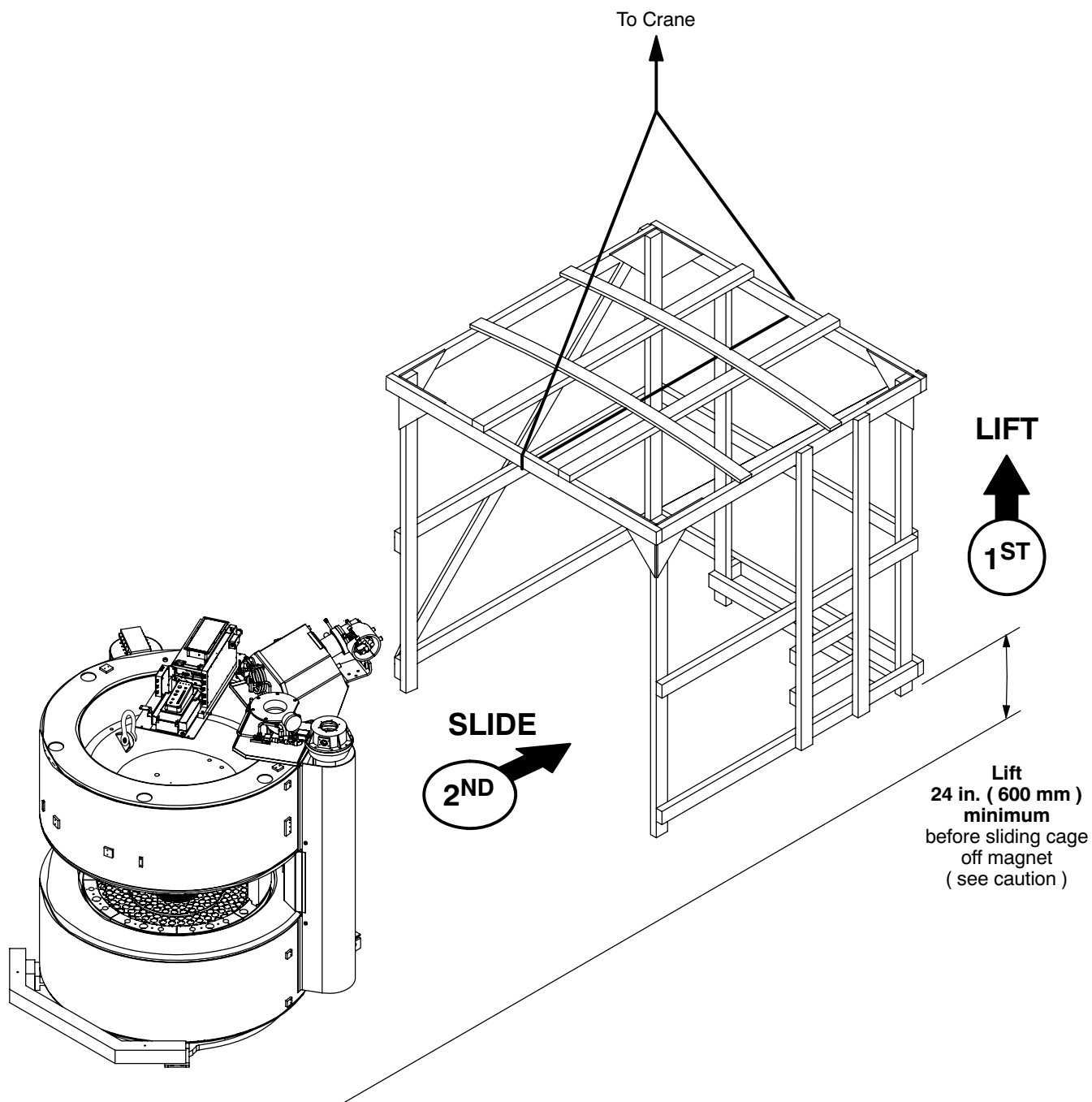
3-4 CAGE REMOVAL (continued)**PROTECTIVE CAGE SIDE REMOVAL**

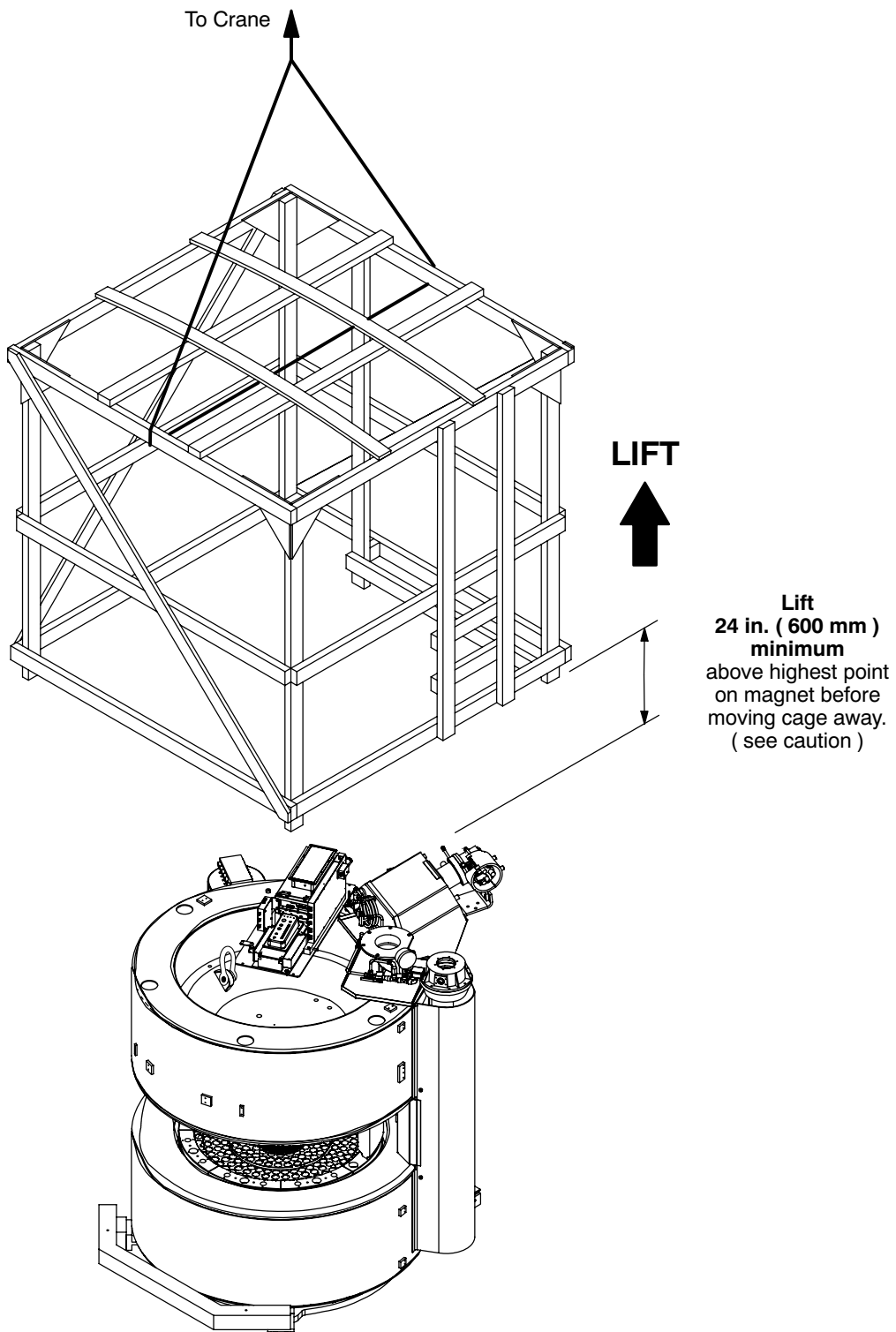
ILLUSTRATION 3-2



3-4 CAGE REMOVAL (continued)

PROTECTIVE CAGE REMOVAL WITH SIDE REMOVED
ILLUSTRATION 3-3



3-4 CAGE REMOVAL (continued)

PROTECTIVE CAGE REMOVAL VERTICALLY
ILLUSTRATION 3-1



■ 3-5 UNLOADING / LIFTING MAGNET



Use caution while cutting holes in the plastic bag (wrap) to avoid serious damage to the magnet components.

1. Cut holes in the protective plastic bag (wrap) for access to the lifting shackles. Be careful not to damage magnet components with knife or by pulling on wrap.

Note

Do not remove the protective plastic bag / wrap surrounding magnet until required for installation purposes inside the Magnet Room.

2. Position crane hook / spreader beam centrally over magnet for a vertical lift. See Illustration 3-2.
3. Attach straps / cables to lifting shackles on top of magnet, spreader beam and crane in conformance with Illustrations 3-2 and 3-3.



Carefully control magnet while lifting, moving or lowering. Some rotation / tilting will occur. Lower smoothly onto a flat surface. Avoid impacts, jolts or excessive tilting (as specified in Table 3-1) as magnet damage may result.

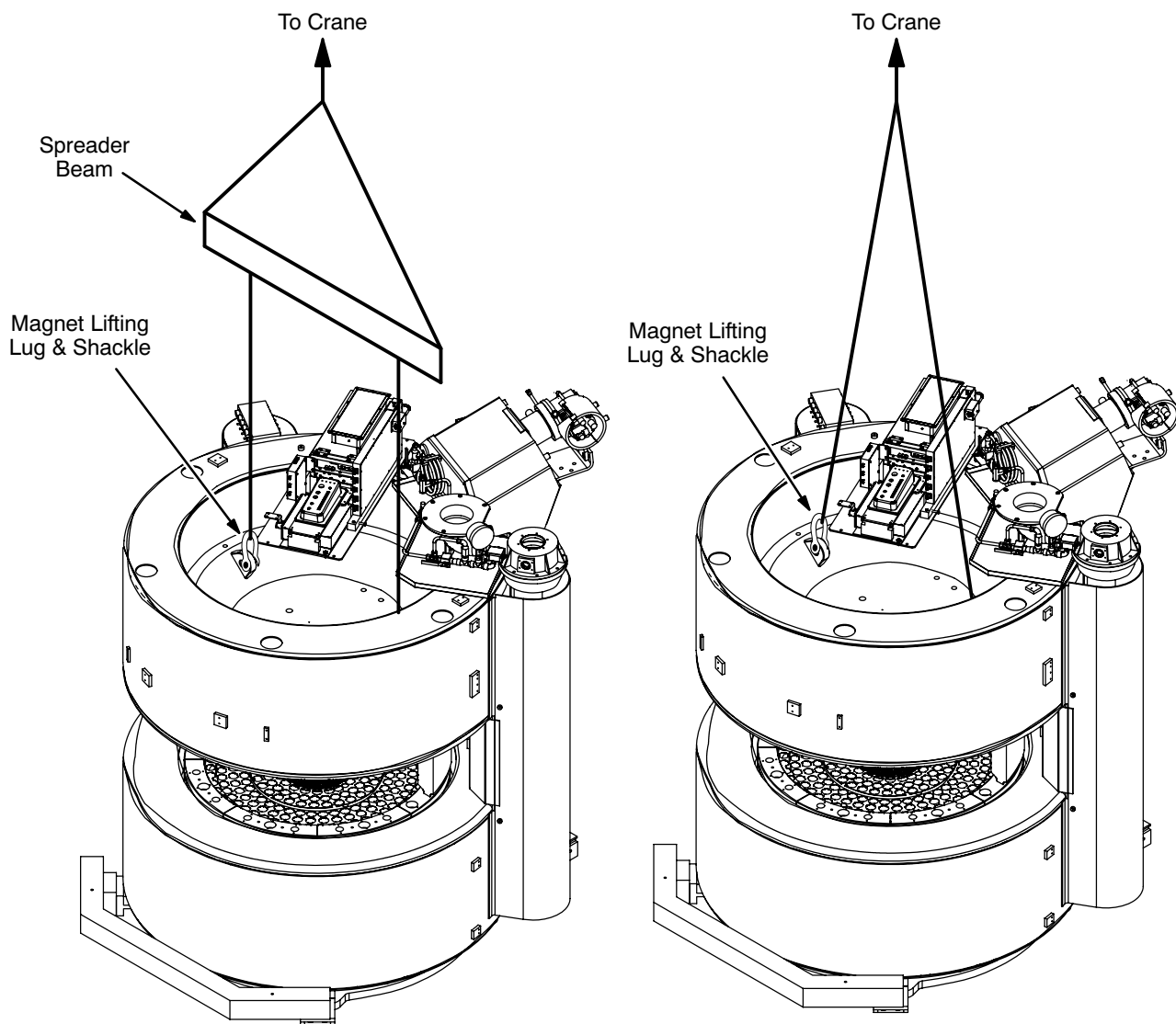
Note

Magnet must be lifted in conformance with requirements shown in Illustrations 3-2 and 3-3 to avoid damage to external components from lifting straps / chains.

4. Lift magnet. Make sure magnet is lowered onto a flat smooth surface.



3-5 UNLOADING / LIFTING MAGNET (continued)



NOTE: 1. 35.50 In. (901.7 mm) center to center between lifting lug holes.

2. DO NOT angle lifting straps / cables outwards from magnet as component damage may result.

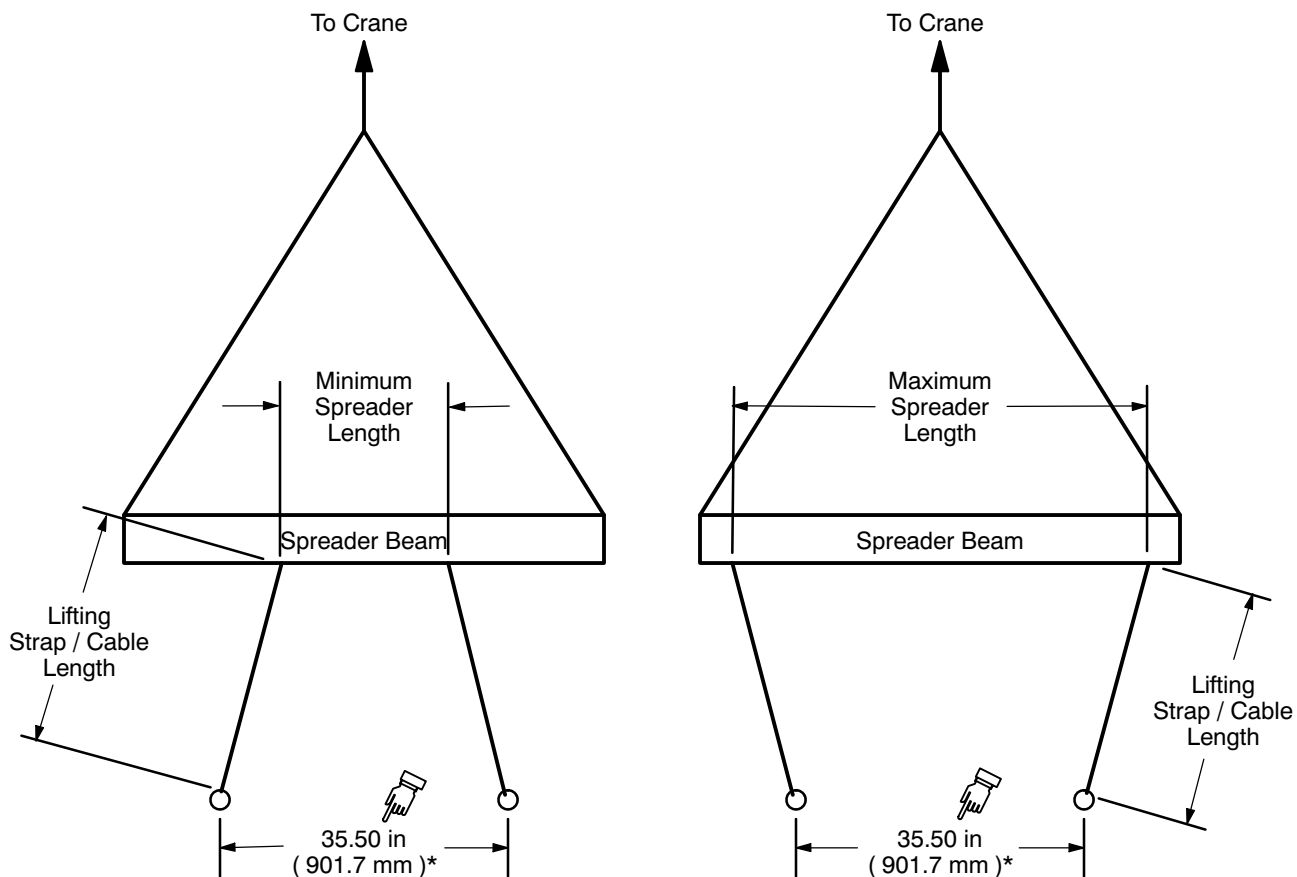
3. See the table in Illustration 3-3 for lifting strap / cable lengths required to avoid damage to magnet components.

CRANE LIFTING MAGNET

ILLUSTRATION 3-2



3-5 UNLOADING / LIFTING MAGNET (continued)



* Distance between Lifting lugs on magnet.

Lifting Strap / Cable Length	Spreader Beam Length	
	Minimum	Maximum
36.0 in. (920 mm)	18.0 in. (460 mm)	56.0 in. (1420 mm)
48.0 in. (1220 mm)	12.0 in. (305 mm)	62.0 in. (1575 mm)
60.0 in. (1525 mm)	6.0 in. (155 mm)	68.0 in. (1725 mm)
72.0 in. (1830 mm)	0**	74.0 in. (1875 mm)
** Use 72.0 in. (1830 mm) length for single point lifting.		

SPREADER BEAM / CABLE LENGTH REQUIREMENTS

ILLUSTRATION 3-3

